

ANALOG & DIGITAL CIRCUITS

Lecture 4

Computer Science Department, TUCN

Digital Circuits

Parameters of integrated logic circuits

- Static transfer characteristic
- Noise margins
- Fan-out & fan-in
- Propagation time
- Power Dissipation

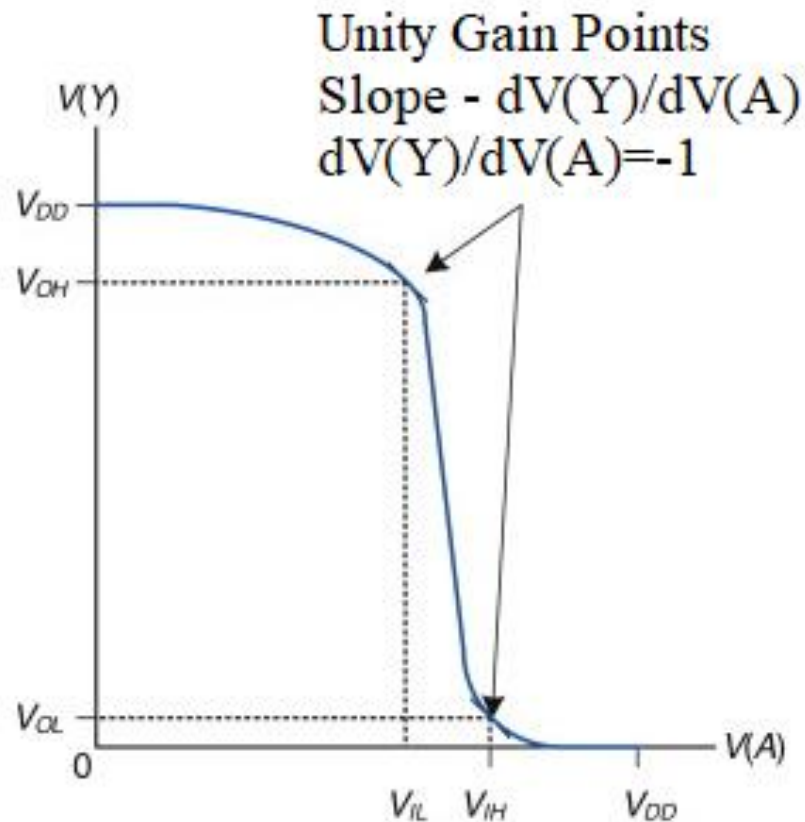
TTL logic circuits

Static Transfer Characteristic

- Output voltage variation function of input dc voltage
- Can't define a unique voltage value for logic '1' or '0'
- Transfer characteristic isn't unique
 - characteristic bounded by two limit curves
 - to each input or output variable, two voltage intervals (domains) will be associated: allowed, and respectively guaranteed

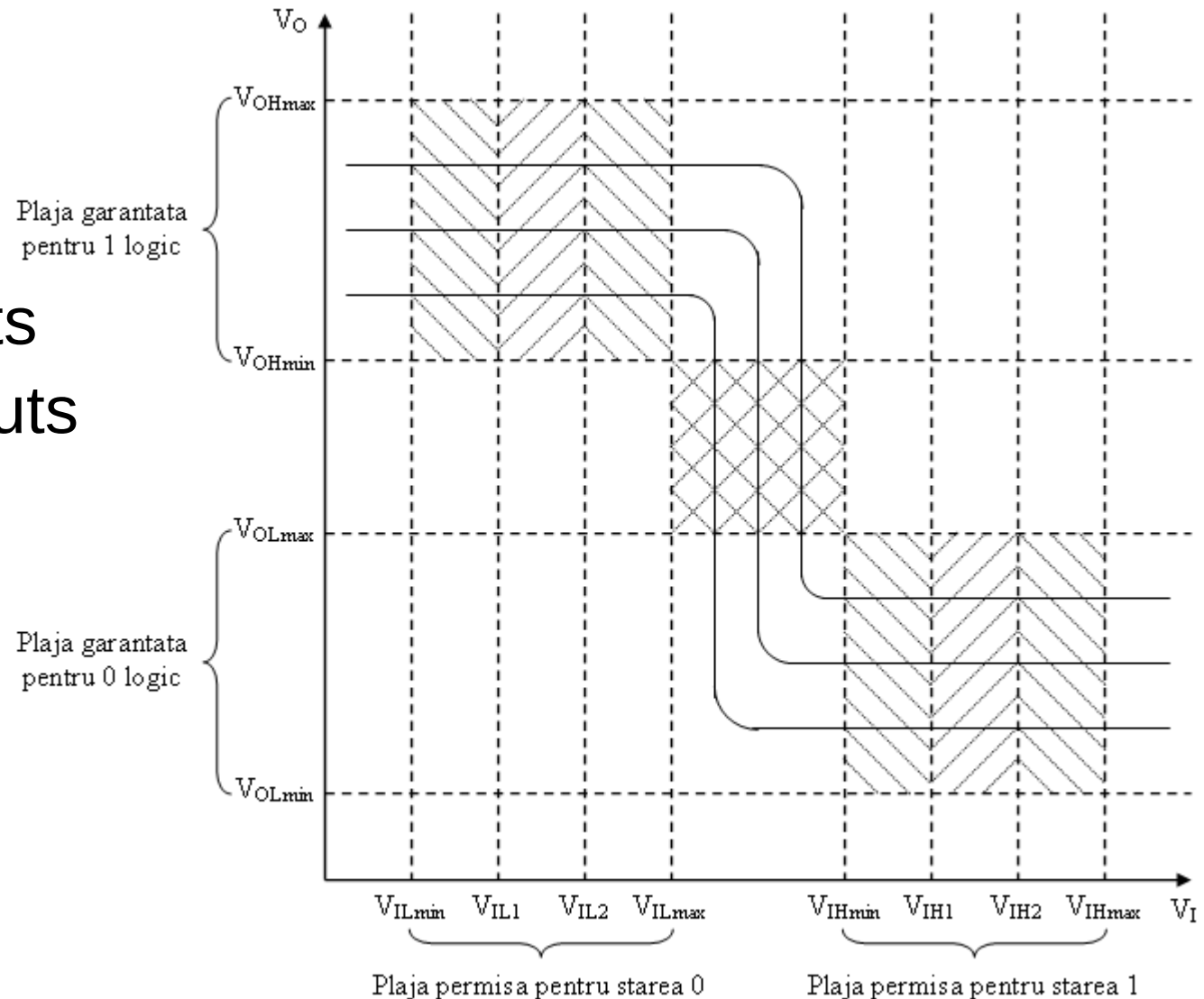
Static Transfer Characteristic

- Output voltage variation function of input dc voltage – inverter example



Static Transfer Characteristic for an Inverter

- Four voltage ranges
- Two for inputs
- Two for outputs
- Defined by eight voltage values



Input/Output voltage values

- V_{ILmin} – minimum voltage level for logic '0' at input
- V_{ILmax} - maximum voltage level for logic '0' at input
- V_{IHmin} - minimum voltage level for logic '1' at input
- V_{IHmax} - maximum voltage level for logic '1' at input
- V_{OLmin} - minimum voltage level for logic '0' at output
- V_{OLmax} - maximum voltage level for logic '0' at output
- V_{OHmin} - minimum voltage level for logic '1' at output
- V_{OHmax} - maximum voltage level for logic '1' at output

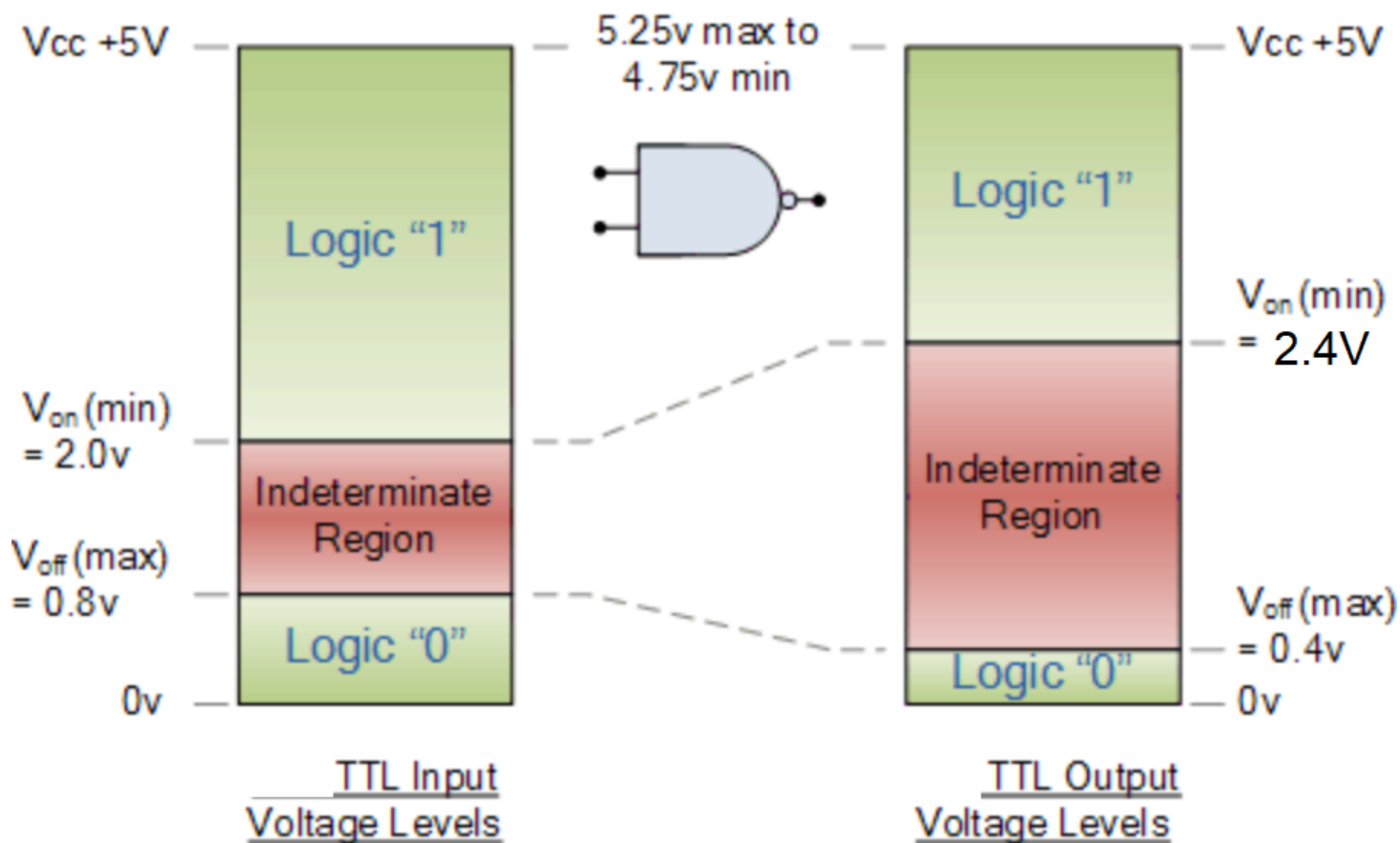
Index meaning

I – *input*

O – *output*

L – *low* logic '0'

H - *high* logic '1'

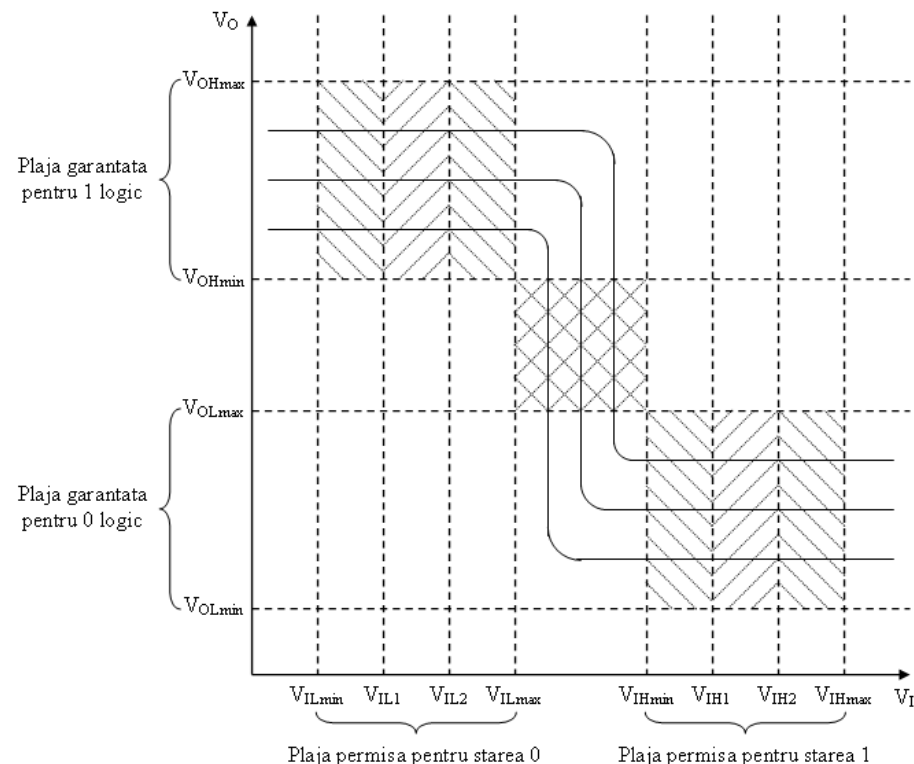


Defining the working regions

- A logic circuit will *work properly* if the input voltage levels will have admitted values (*work in allowed regions*)
- Obtained output voltage levels will be into guaranteed regions
- Considering that output voltage will become input voltage for driven circuit(s), have five working regions:
 - Normal working region for logic '0' and logic '1'
 - Working region in presence of noise signals for logic '0' and respectively '1'
 - Transitory region

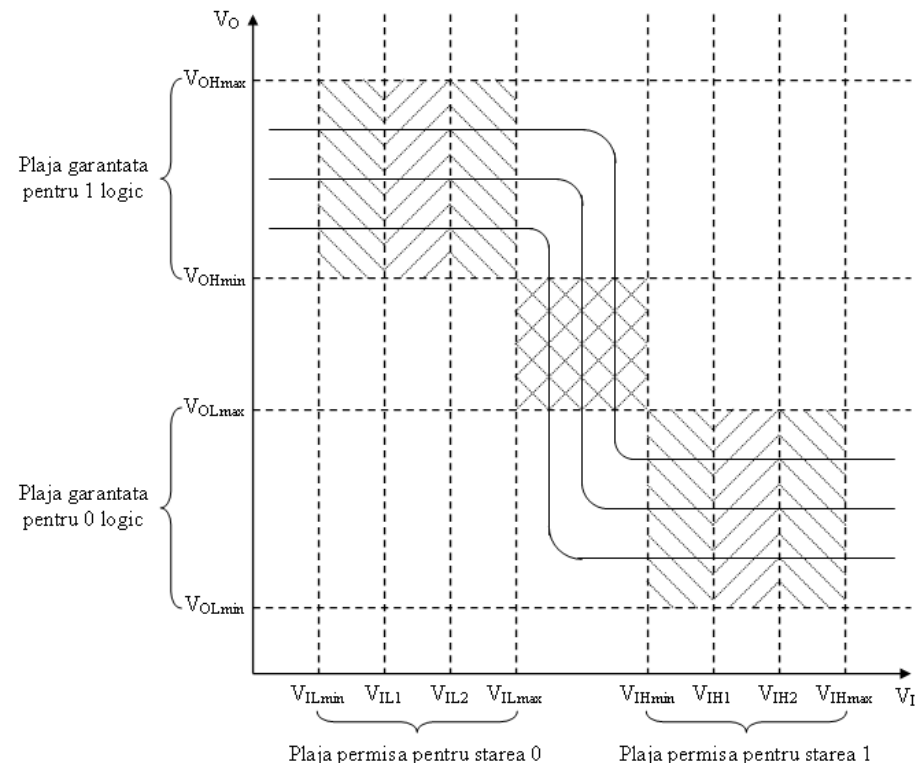
Normal working region (no noise signals)

- For low input voltage level (L), between values $V_{IL2} - V_{IL1}$
- For high input voltage level (H) between values $V_{IH2} - V_{IH1}$



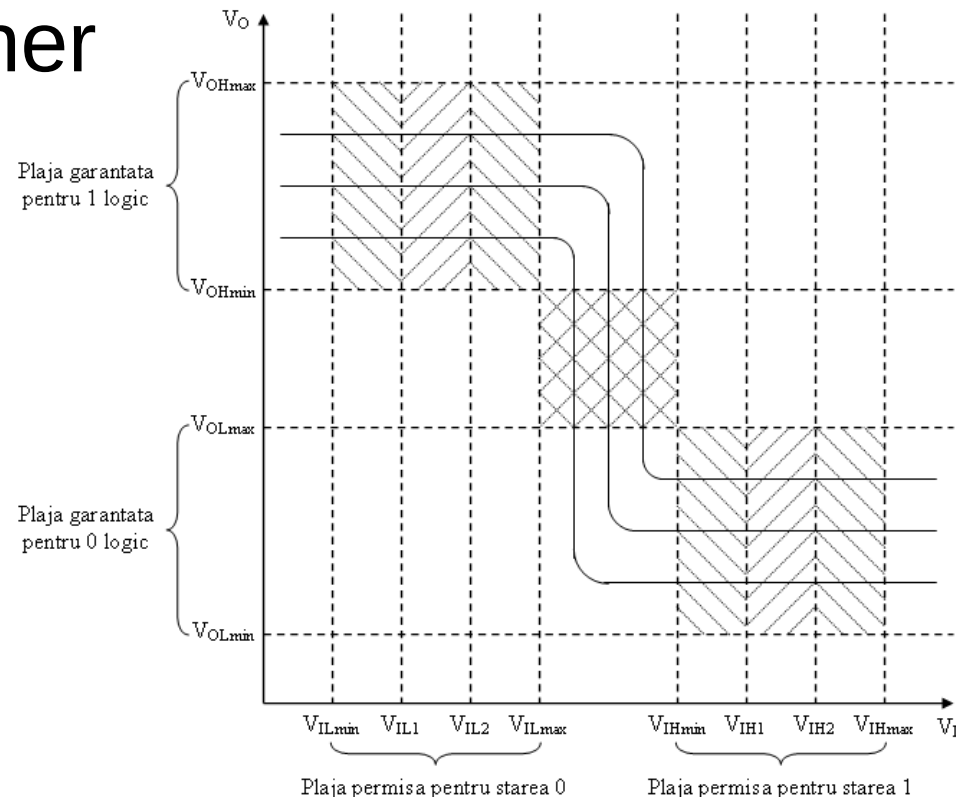
Working region when noise signals apply

- For low input voltage level (L), between values $V_{ILmax} - V_{ILmin}$
- For high input voltage level (H) between values $V_{IHmax} - V_{IHmin}$



Transitory region

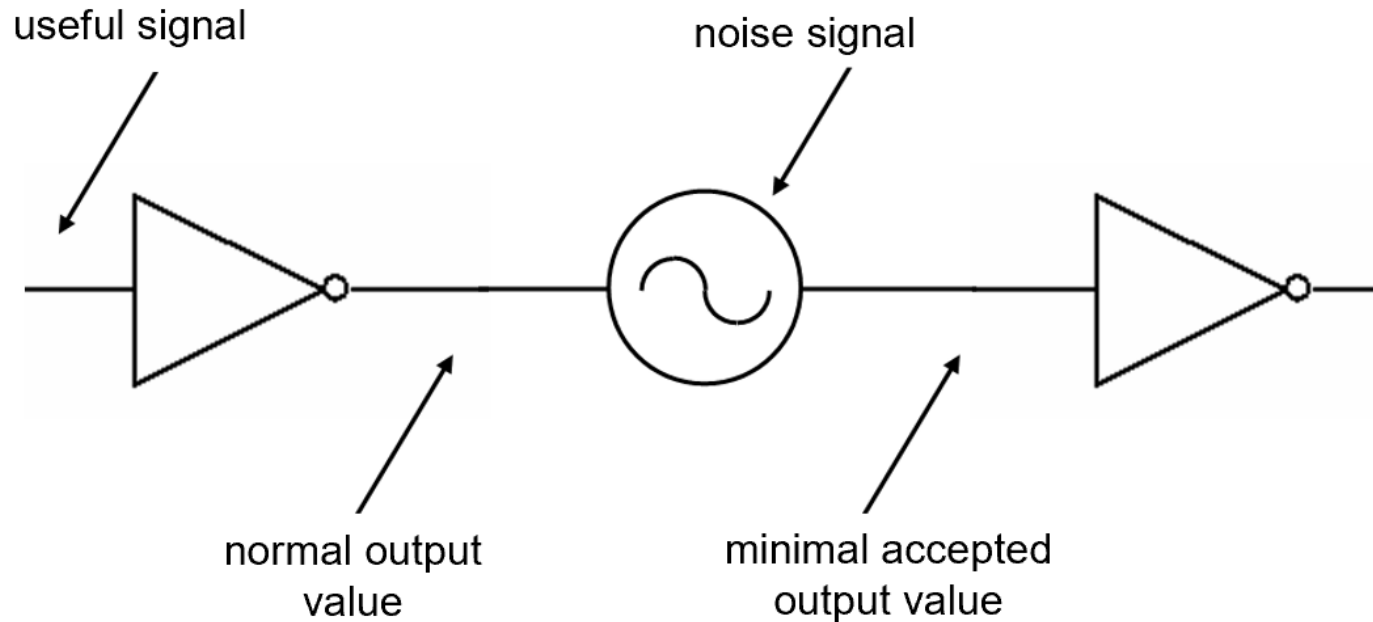
- For input voltage levels between V_{IHmin} – V_{ILmax}
- Those input voltages driving circuit from one stable state to the other



Noise margins

- Noise (immunity) margins: stability at static perturbations
- the noise margin is the peak amount of spurious or "noise" voltage that may be superimposed on a weak gate output voltage signal before the receiving gate might interpret it wrongly
- Guaranteed noise margin for a logic state is given by difference between guaranteed output voltage level of the driving circuit and the worst case for the input voltage level accepted for that state by the driven circuit

Guaranteed noise margins (manufacturer's specifications)

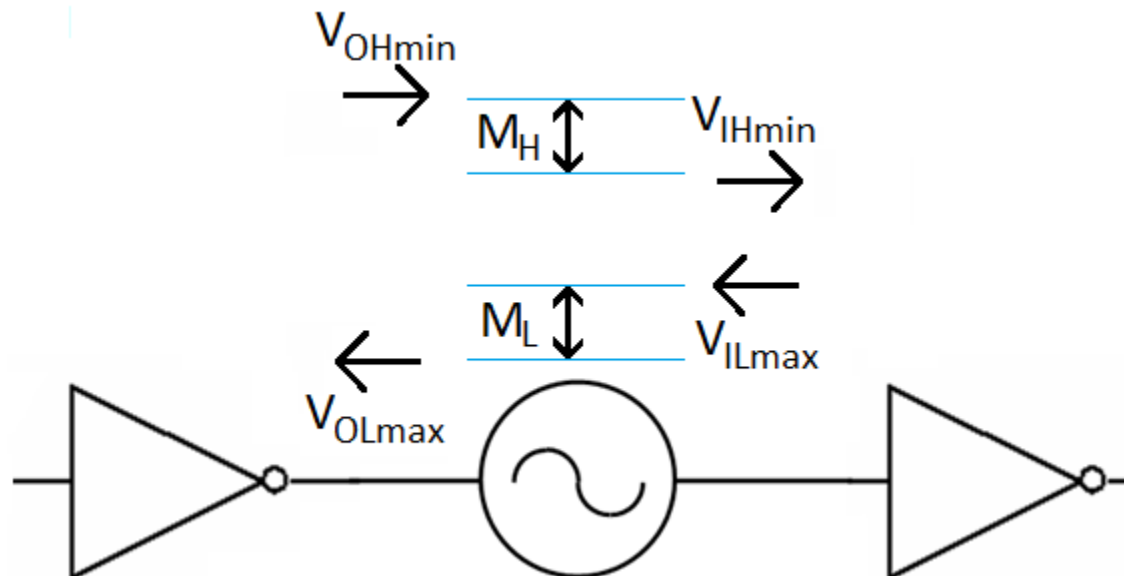
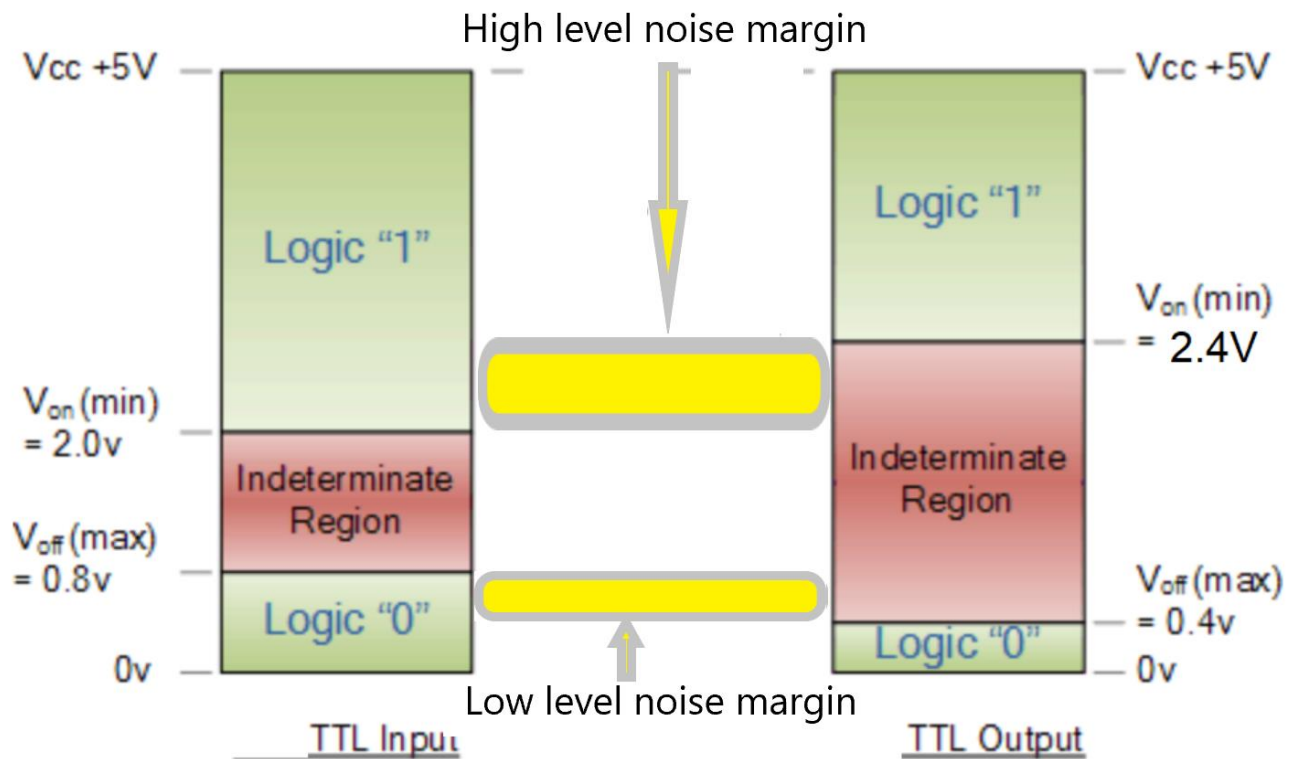


- For logic '0' state:

$$M_L = V_{ILmax} - V_{OLmax}$$

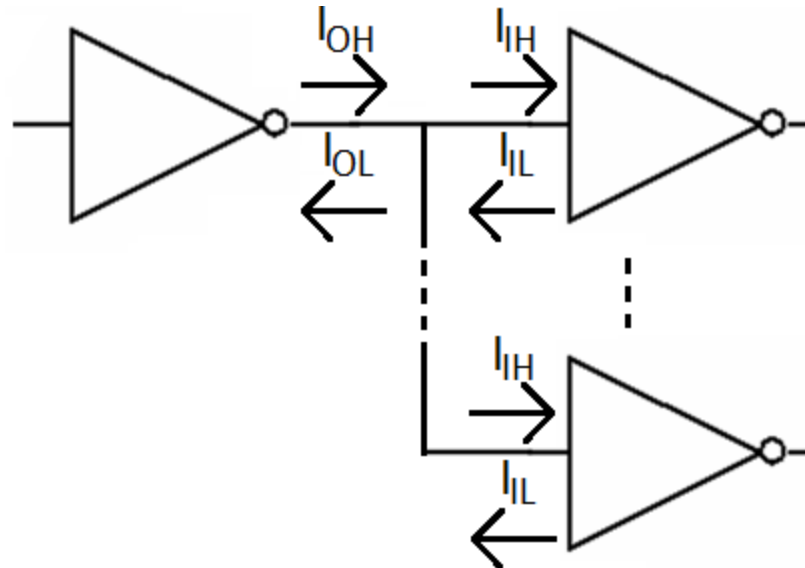
- For logic '1' state:

$$M_H = V_{OHmin} - V_{IHmin}$$



Fan-out and Fan-in

- The input of a circuit is a load for its driver circuit
- For a logic circuit generates guaranteed output voltage levels is a must to be driven at inputs with corresponding current levels
- For circuit interconnection, important to design the output current of the driving circuit, and to consider the sum of input currents of the driven circuits

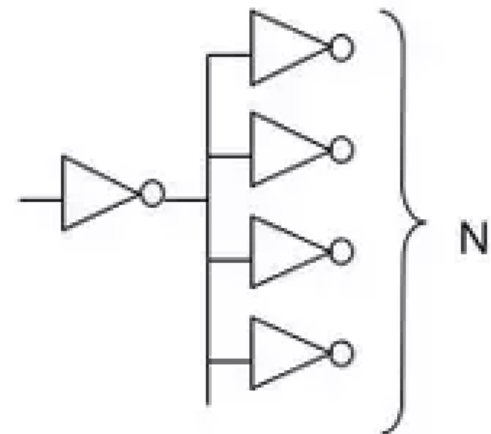


Fan-in factor for logic circuits

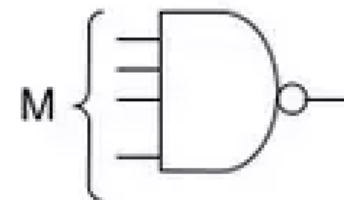
- Fan-in (FI) and Fan-out (FO) factors are defined based on some current values, those corresponding to allowed input voltages and respectively to guaranteed output voltages, in the worst case: I_{ILmax} , I_{IHmin} , I_{OLmax} și I_{OHmin}
- For any integrated circuits family, a basic (fundamental) gate is defined, and the fan-in/ fan-out values for the rest of circuits are defined as multiples of the values for that basic gate
- For an input, the FI value means the number N ($N > 1$) of standard inputs (i.e. of basic gate) equivalent to that input: $FI = N$

Fan-Out and Fan-In

- Fan-out – number of load gates connected to the output of the driving gate
 - gates with large fan-out are slower



- Fan-in – the number of inputs to the gate
 - gates with large fan-in are bigger and slower



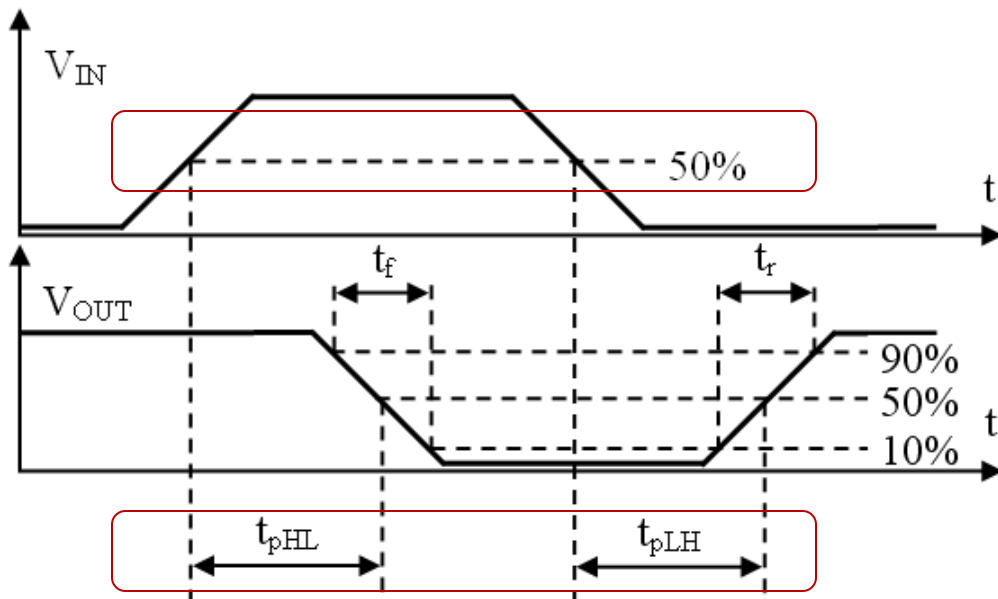
Fan-out factor for logic circuits

$$FO_L = \left\lfloor \frac{|I_{OL}|}{|I_{IL}|} \right\rfloor, FO_H = \left\lfloor \frac{|I_{OH}|}{|I_{IH}|} \right\rfloor, FO = \min(FO_L, FO_H)$$

- When interconnecting logic circuits (from a family), the following relations must be satisfied (associate with the worst case):

$$I_{OL} \geq \sum_n^{i=1} I_{IL}, \quad I_{OH} \geq \sum_n^{i=1} I_{IH}$$

Propagation Time



- Raising-up and fall-down times (t_r , t_f) are defined using ratios of signal amplitude (0.1 and 0.9)

- Propagation times (delay) (t_{pHL} si t_{pLH}) are defined for halves the input/output signal amplitudes
- Average propagation time: $t_{pd} = (t_{pHL} + t_{pLH}) / 2$
- Important parameter for any circuit, giving sign of performance

Power Dissipation

- Parameter depending on:
 - Power supply voltage (V_{CC});
 - Absorbed currents from V_{CC} when output is logic '1' (I_{CCH}), or '0' logic (I_{CCL});
 - Output current on shortcircuit (I_{OS});
 - Average power consumption (P_m);

$$P_m = P_{CC} + P_c = \frac{I_{CCH} + I_{CCL}}{2} V_{CC} + f C_P V_{CC}^2$$

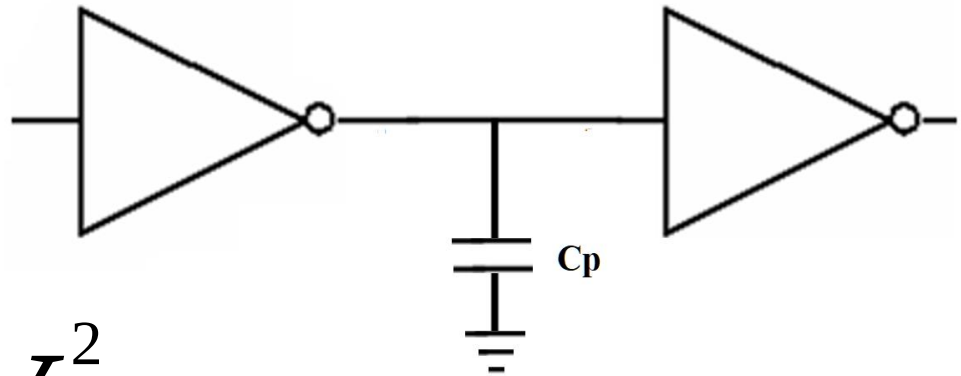
P_{CC} – small values; P_c – higher values

Average power dissipation on cc

$$P_{CC} = \frac{P_H + P_L}{2} = \frac{I_{CCH} + I_{CCL}}{2} \cdot V_{CC}$$

Power dissipation on ac

- Important power component, due to charging/discharging of **stray output capacitances C_p**
- Power consumption during the switching regime:



$$P_C = f C_P V_{CC}^2$$

f – switching frequency

Total power dissipation

$$P_m = P_{CC} + P_c = \frac{I_{CCH} + I_{CCL}}{2} V_{CC} + f C_P V_{CC}^2$$

TTL Logic Integrated Circuits

- General considerations
- TTL standard series
 - TTL basic (fundamental) gate
 - Circuit description
 - Gate operation
 - Parameters of TTL basic gate



A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

General Considerations

- TTL (Transistor-Transistor-Logic)
- Family with a lot of circuit series, developed based on a trade-off between propagation speed and power dissipation
- Standard, high-speed (H), Low power (L), Schottky (S)

Input stage

- A and B

Driver transistor

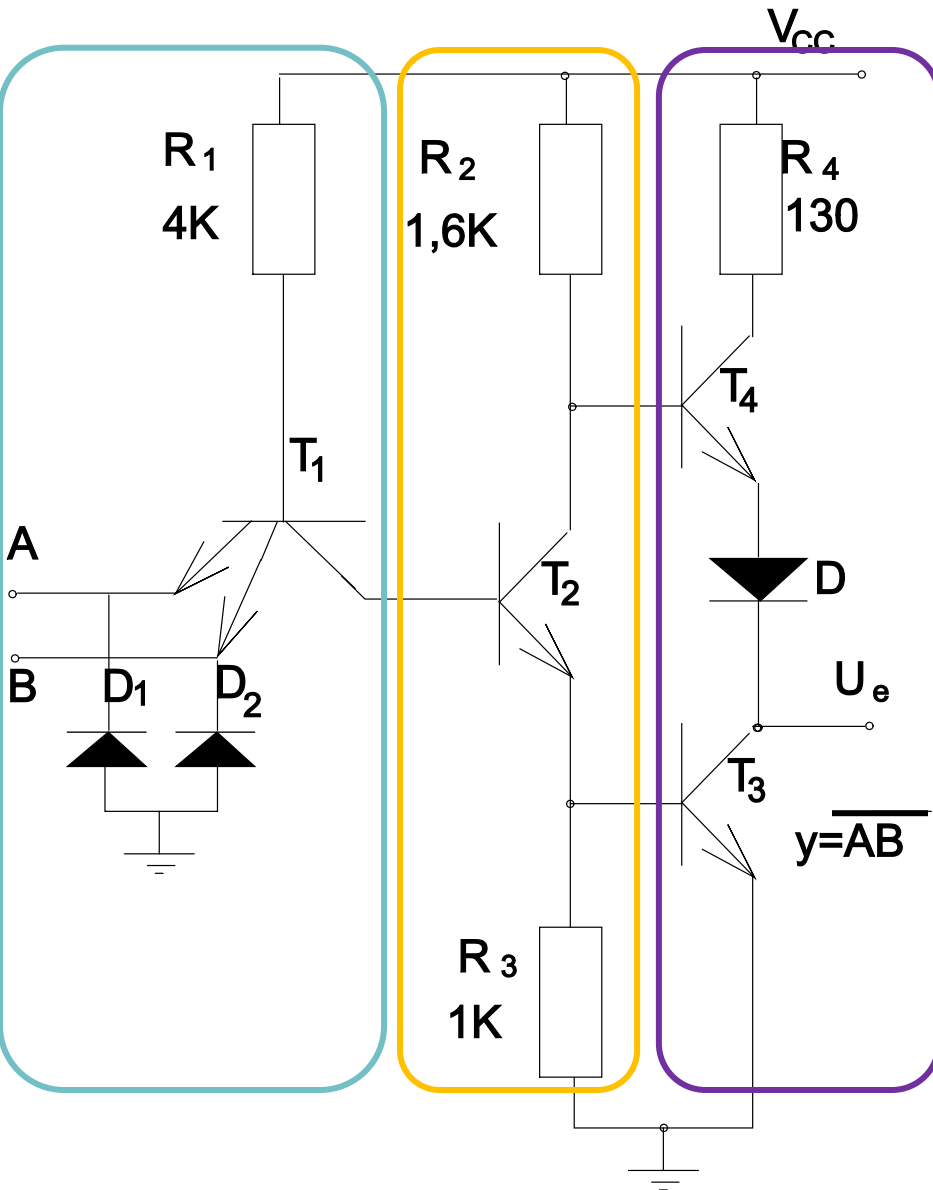
Output stage

$$U_e = \overline{A * B}$$

The term “**drive**” in electronics means:

- give the **appropriate** amount of power (voltage and current) to a device.
- control another circuit or component

TTL Basic Gate



Input stage

- Multi-emitter transistor T_1
- Clipping diodes D_1 , D_2

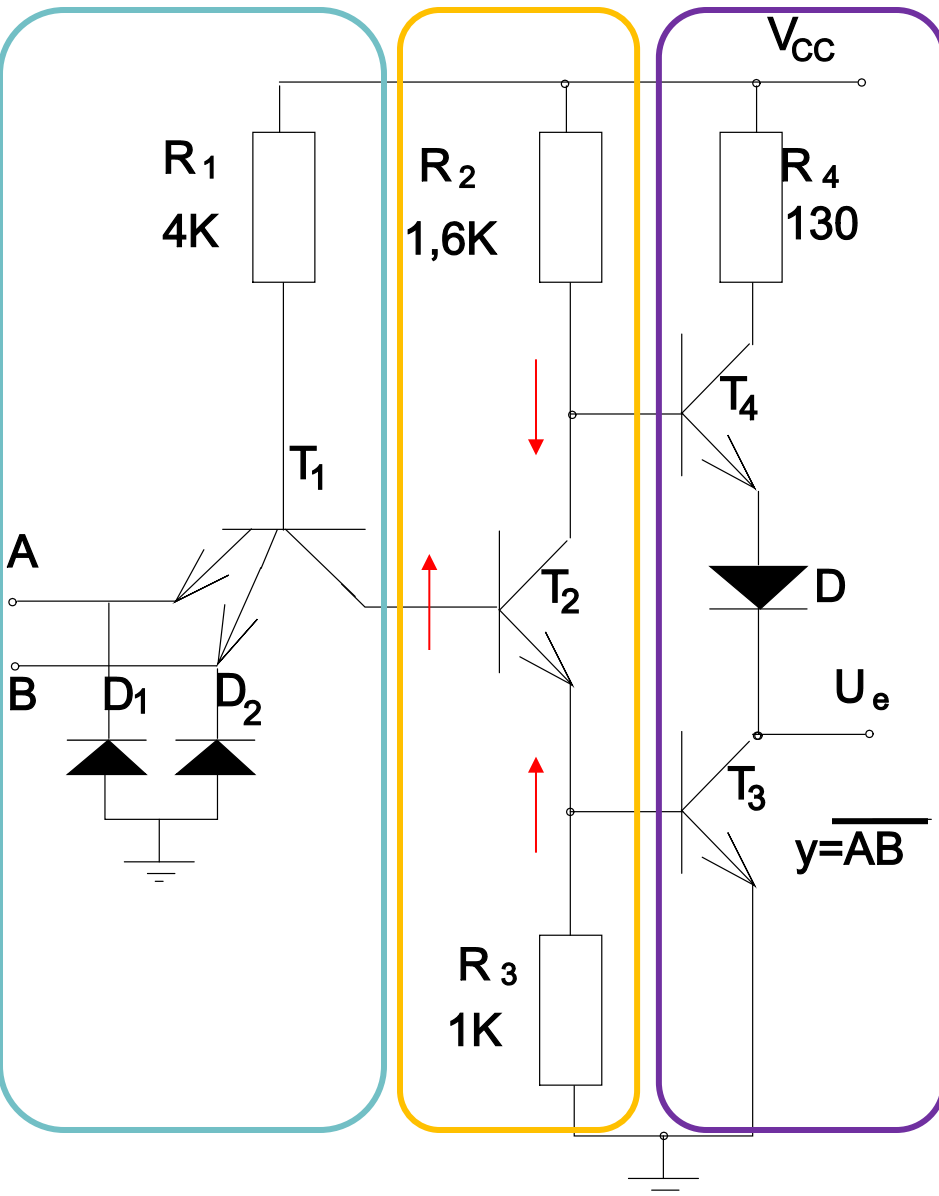
Driver transistor

- Transistor T_2

Output stage (totem-pole arrangement)

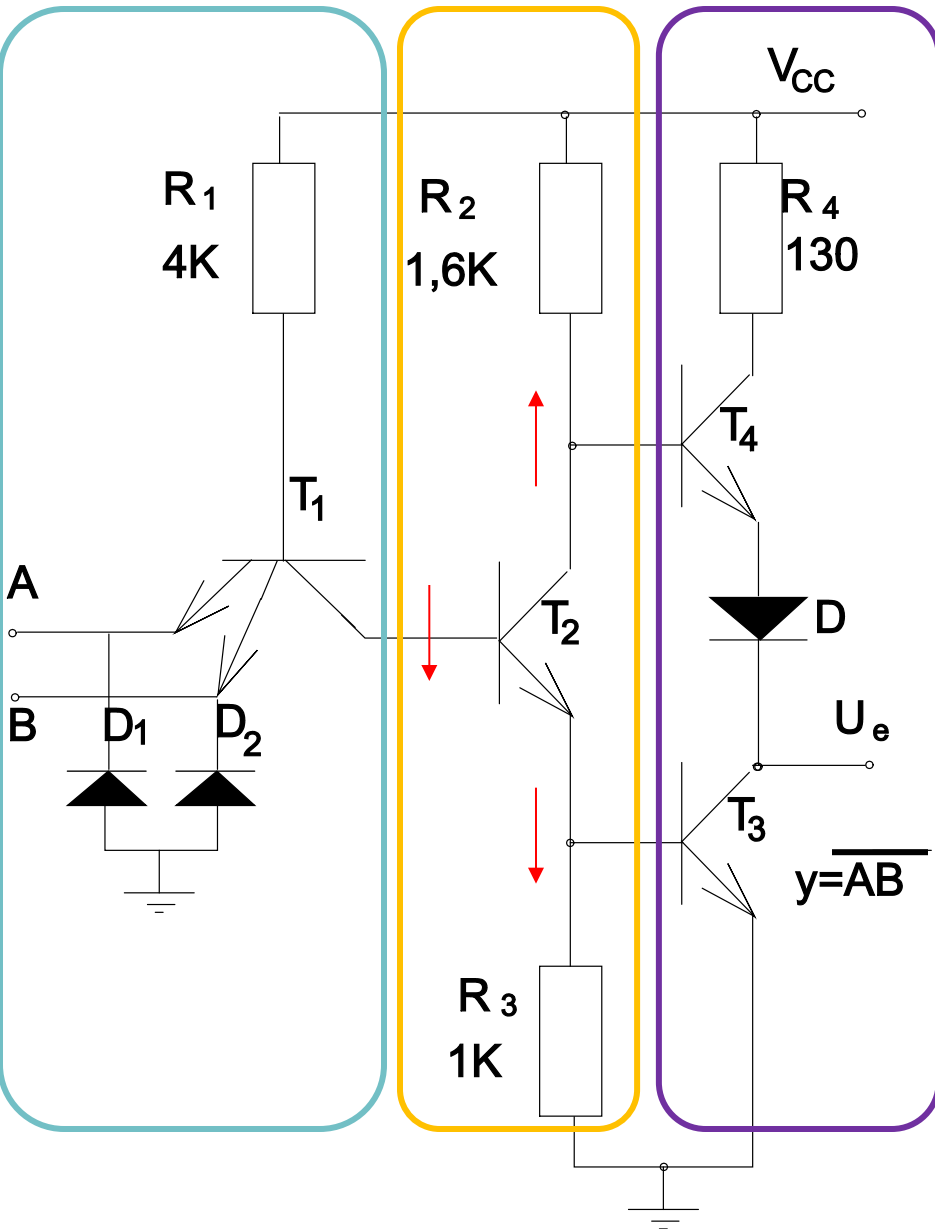
- transistors T_4 și T_3
- diode D

Gate operation when both inputs at logic '1'



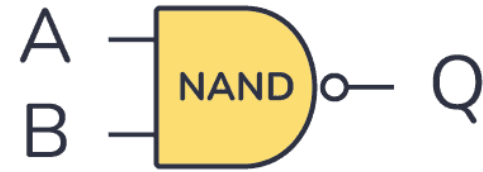
- T_1 base-emitter junctions reverse biased (reverse active region)
- T_1 base-collector junction and base-emitter junctions of T_2 & T_3 make a chain of open diodes (forward biased by R_1 from power supply), T_2 & T_3 saturated
- T_4 off due to base potential, lower than emitter's, due to presence of diode D
- $U_e = U_{CES(T_3)}$ corresponding to logic "0"

Gate operation for one input '0'



- T₁ saturated, voltage from T₁ collector lowers, transistor T₂ off
- Low voltage level from T₂ emitter drives T₃ off
- High potential of T₂ collector opens transistor T₄
- U_{R2} low, U_{BE(T4)} + U_D ≈ 1,5V, U_e > 3,4V corresponding to logic level "1"

$$U_e = \overline{A * B}$$

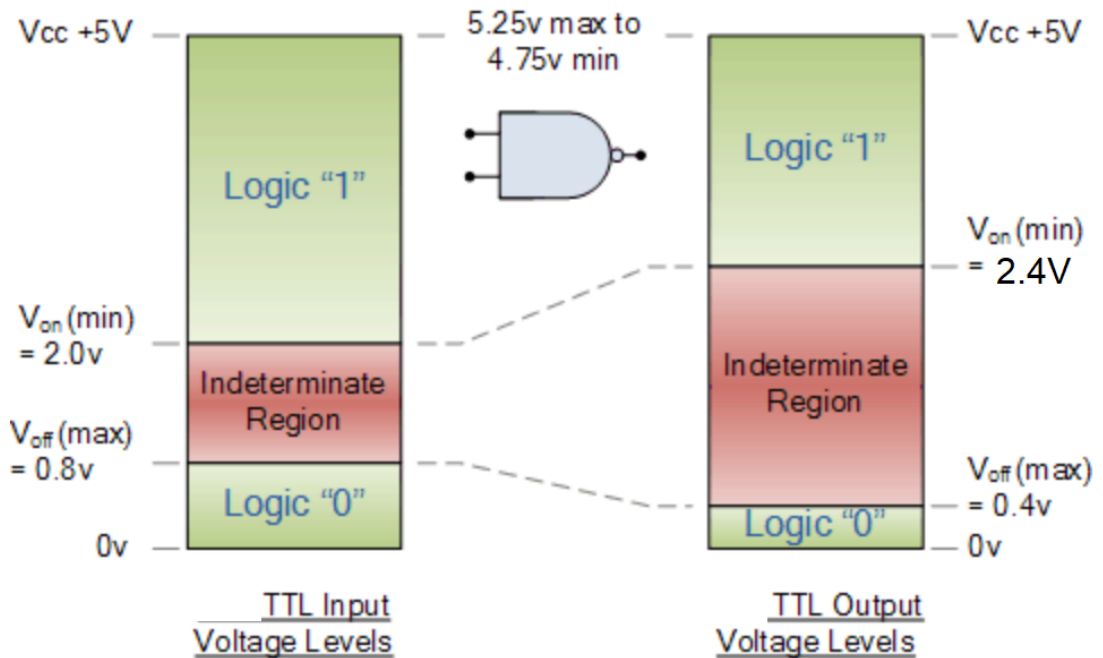


A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

- Transistors T_4 & T_3 switch in counter-time, making R_4 *being low* (130Ω), building a low output impedance and a small time constant for charge/discharge of output stray capacitances

Logic Levels

- $V_{ILmax} = 0.8 \text{ V}$
- $V_{IHmin} = 2 \text{ V}$
- $V_{OLmax} = 0.4 \text{ V}$
- $V_{OHmin} = 2.4 \text{ V}$
- $V_T = 1.3\text{V}$, threshold voltage, same value for input and output voltages



Noise Margins

- Guaranteed values

$$M_L = V_{ILmax} - V_{OLmax} = 0.8V - 0.4V = 0.4V$$

$$M_H = V_{OHmin} - V_{IHmin} = 2.4V - 2V = 0.4V$$

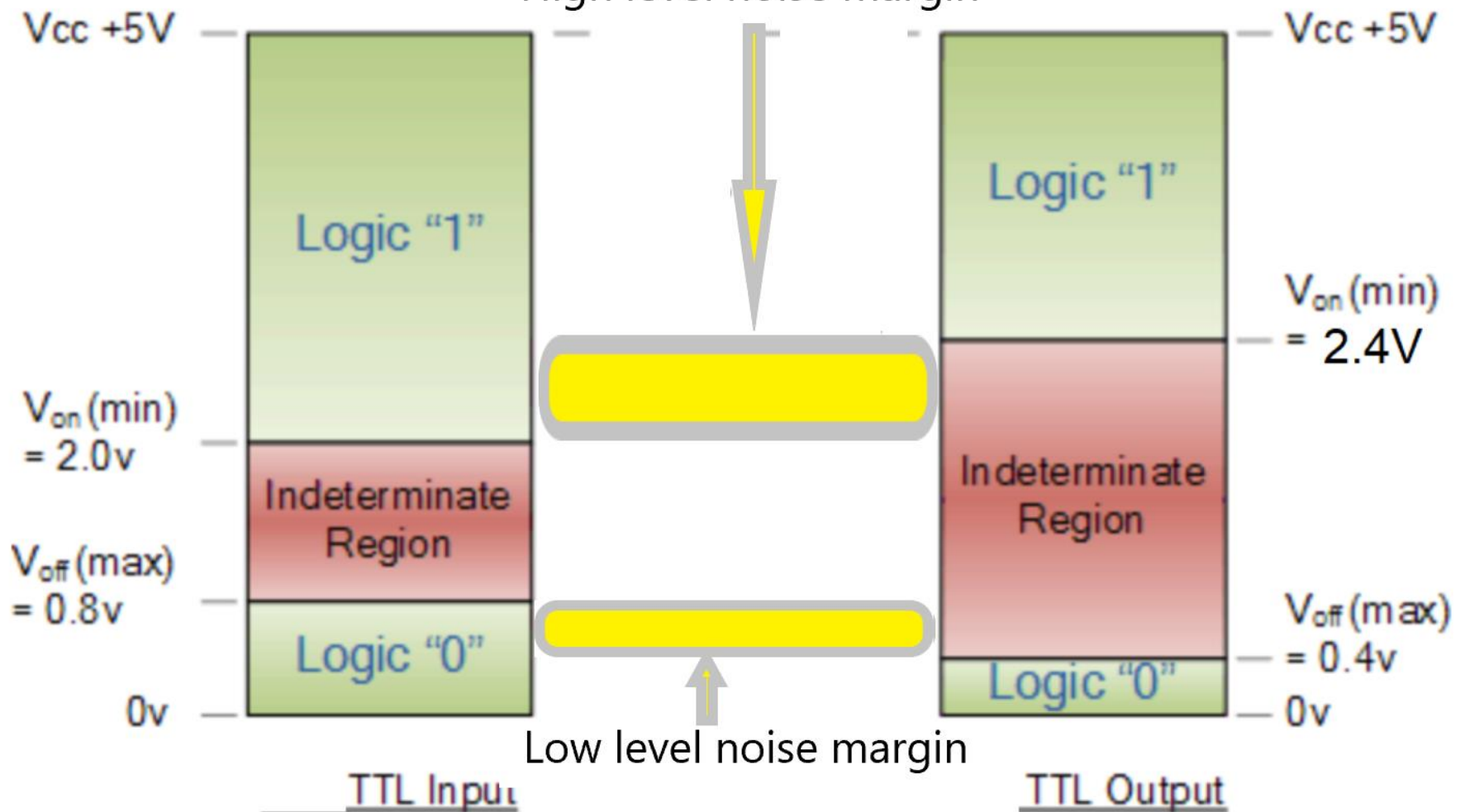
- Real values

$$M_L = V_T - V_{OL} = 1.3V - 0.2V = 1.1V$$

$$M_H = V_{OH} - V_T = 3.5V - 1.3V = 2.2V$$

- It implies that preferred output idle state being '1' logic, and switching command being 'zero active', i.e. a signal going from high to low

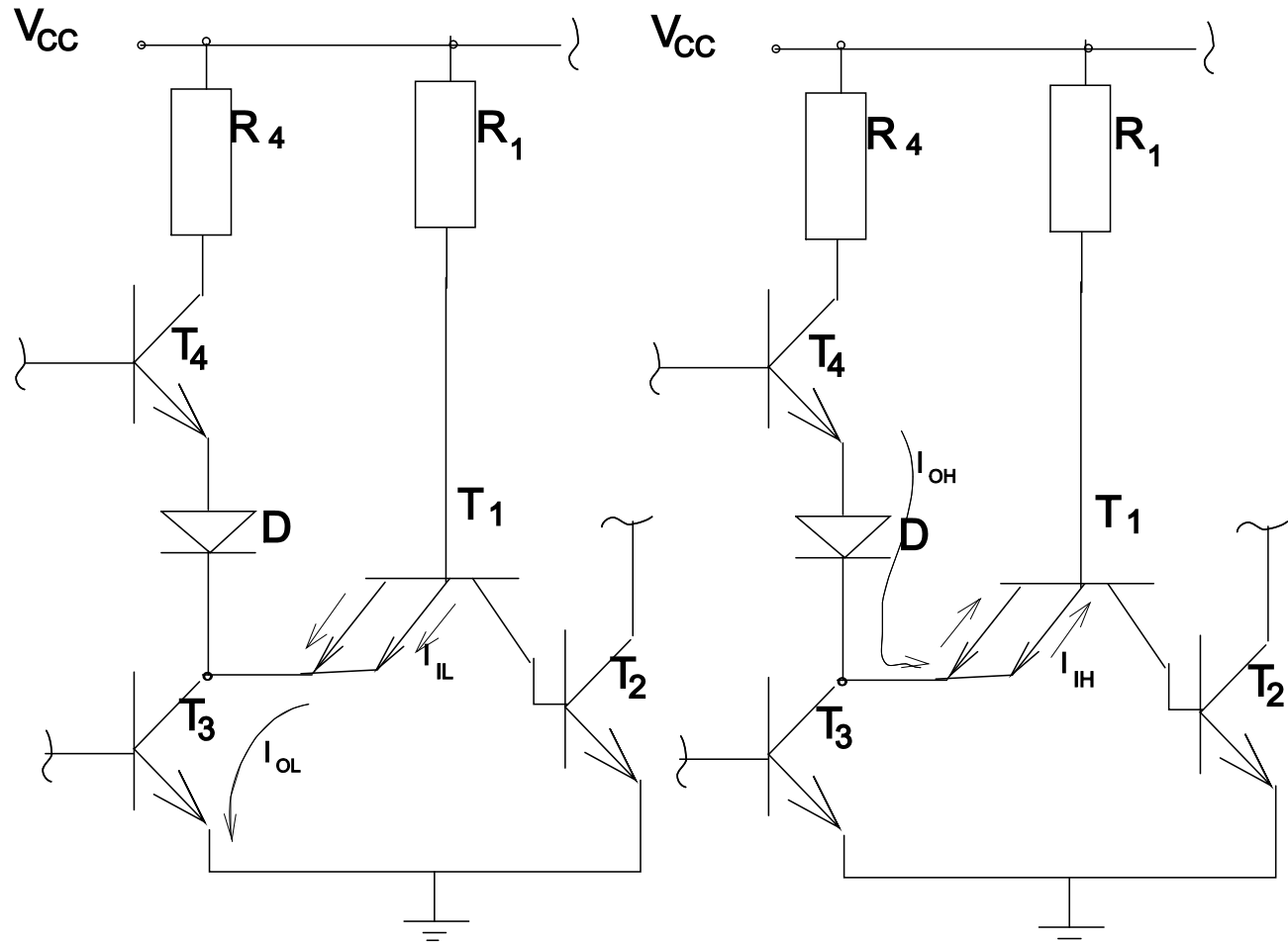
High level noise margin



Input & output currents

- By convention: positive value if gate sinks current and negative value if gate generates currents
- $I_{IH} = 40 \mu\text{A}$
- $I_{IL} = -1,6 \text{ mA}$
- $I_{OH} = -800 \mu\text{A}$
- $I_{OL} = 16 \text{ mA}$

Fan-in/ fan-out



$$FI_L = 1, I_{IL} = -1.6\text{mA}$$

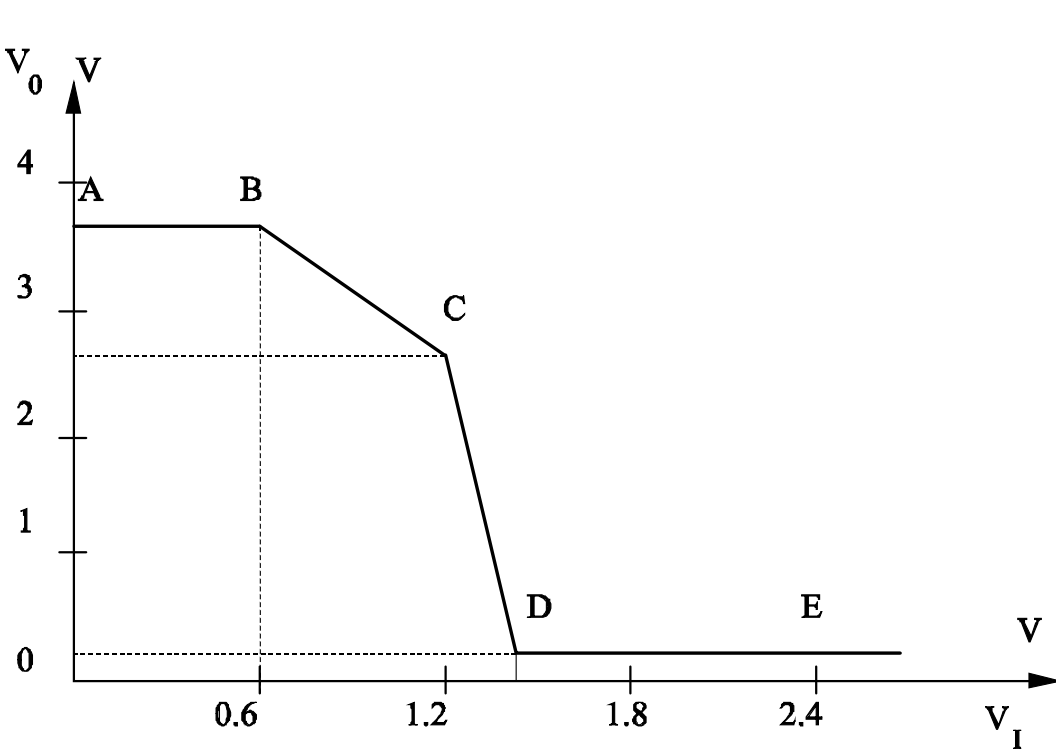
$$FI_H = 1, I_{IH} = 40\mu\text{A}$$

$$FO_L = \left\lfloor \frac{|I_{OL}|}{|I_{IL}|} \right\rfloor = \left\lfloor \frac{16\text{mA}}{1.6\text{mA}} \right\rfloor = 10, FO_H = \left\lfloor \frac{800\mu\text{A}}{40\mu\text{A}} \right\rfloor = 20, FO = \min(FO_L, FO_H) = 10$$

Static transfer characteristic

$0V < U_i < 0,65V$, T_1 on, T_2 off, $U_e = V_{CC} - R_2 \cdot I_{R2} - U_{BE(T4)} - V_D$, $U_{BE(T4)} = V_D = 0,75V$, $I_{R2} \approx I_{B(T4)} = I_{OH}/(\beta_N + 1)$, $U_e = 3,4V$, on AB segment.

$0,65V < U_i < 1,3V$, T_2 starts conducting, going into forward active region. Current gain for transistor T_2 over segment BC is: $\alpha \approx -R_2/R_3$. T_4 repeater, T_3 off, state for segment BC.



$1,3V < U_i < 1,5V$, T_3 starts conducting, U_e lowers quickly (segment CD). T_2 , T_4 and T_3 conducting into forward active region. Current sink from the power supply increases.

$1,5V < U_i < 2,25V$, T_4 off, T_3 saturated, $U_e = U_{CEs(T3)} \approx 0,2V$, region DE.

Power dissipation

$$P_{CC} = \frac{I_{CCH} + I_{CCL}}{2} V_{CC}$$

$$I_{CCH} = I_{R1} = (V_{CC} - V_{B(T1)}) / R_1 \approx 1\text{mA}$$

$$I_{CCL} = I_{E(T2)} = I_{C(T2)} + I_{B(T2)} = (V_{CC} - V_{C(T2)}) / R_2 + (V_{CC} - V_{B(T1)}) / R_1 \approx 3,3\text{mA}$$

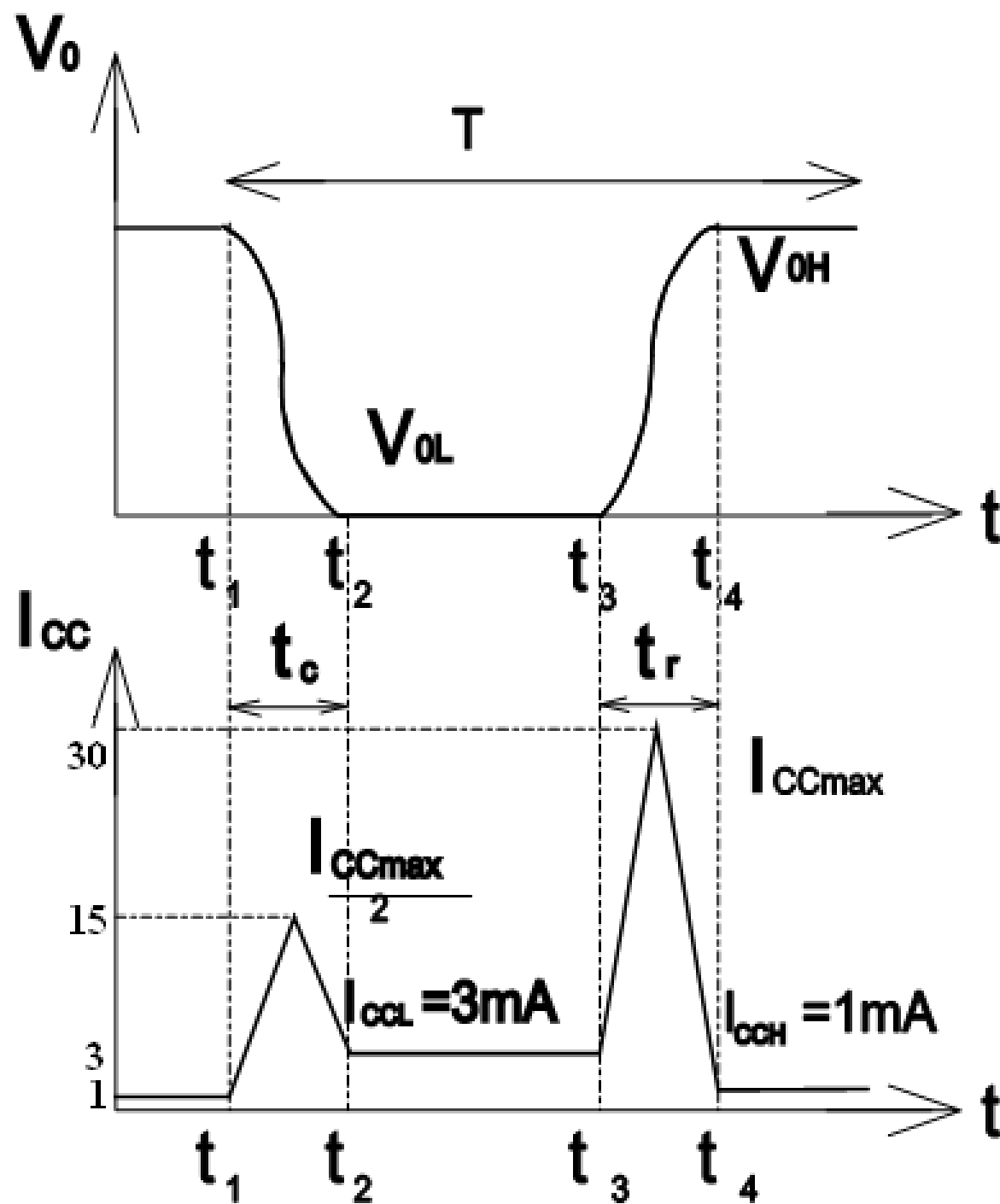
$$P_{CC} \approx 10\text{mW}$$

$$P_C = C_p V_{CC}^2 f$$

$$C_p = 15\text{pF}; f = 1\text{MHz}, P_C \approx 0,4\text{mW}; f = 20\text{MHz}, P_C \approx 7,5\text{mW}$$

- Besides the two components P_{CC} and P_C , there is another, due to simultaneous conduction of transistors T_3 and T_4 . This extra power consumption P_{DS} has the formula:

$$P_{DS} = V_{CC} \left(\frac{I_{CCmax}}{2.2} \cdot \frac{t_c}{T} + \frac{I_{CCmax}}{2} \cdot \frac{t_r}{T} \right)$$



Propagation Delay

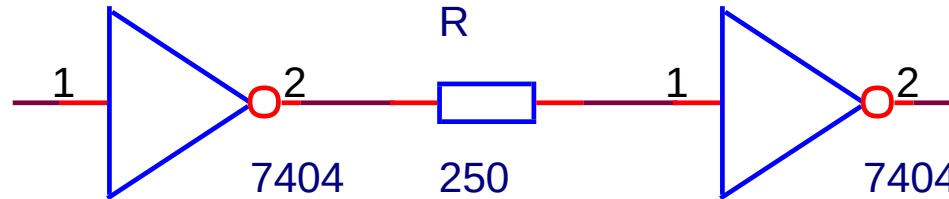
- Given by charging/discharging times of stray capacitance from gate's output and by switching times of transistors
- $t_{pHL} = t_{c1} + t_{des}$
- $t_{pLH} = t_{c2} + t_{inc}$
- $t_{pd} = (t_{pHL} + t_{pLH})/2$
- Switching times: $t_{c1} = 5\text{ns}$ and $t_{c2} = 8\text{ns}$
- Charging/discharging times of stray capacitances:

$$t_{des} = C_p \frac{V_{OH} - V_{OL}}{I_{OL}} \qquad t_{inc} = C_p \frac{V_{OH} - V_{OL}}{I_{OH}}$$

- Formula for t_{inc} , short-circuit current value is considered I_{OS}
- For $I_{OS} = 18\text{mA}$ results $t_{inc} = 2.5\text{ns}$
- Calculated values: $t_{pHL} = 8\text{ns}$ și $t_{pLH} = 10.5\text{ns}$
- Data book values: **$t_{pHL} = 8\text{ns}$** and **$t_{pLH} = 12\text{ns}$** , resulting **$t_{pd} = 10\text{ns}$**

Proposed problems

- Find out the maximum value of a resistor may be connected between two standard TTL gates, without modifying the circuit behavior. How this resistor is affecting the noise margins?



$$R_{L\max} = \frac{V_{IL\max} - V_{OL\max}}{I_{IL}} = 250\Omega$$

$$R_{H\max} = \frac{V_{OH\min} - V_{IH\min}}{I_{IH}} = 10K\Omega$$

$$R_{\max} = 250\Omega$$

- Increasing the resistance causes the noise margin to decrease

- Design a circuit based on a NAND TTL standard gate, able to drive a LED. For the LED, following values are considered: $V_{LED}=1,6V$ and $I_{LED}=10mA$.
- if $I='0'$ -> LED off
- if $I='1'$ -> LED on

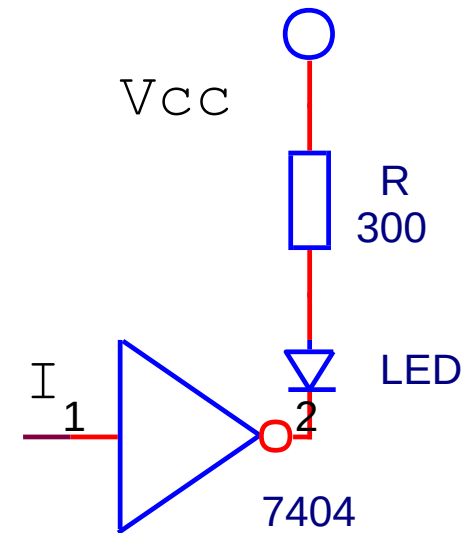
$$R = \frac{V_{CC} - V_{LED} - V_{OLmax}}{I_{LED}} = 300\Omega$$

$$I_{IH} = 40 \mu A$$

$$I_{IL} = -1,6 \text{ mA}$$

$$I_{OH} = -800 \mu A$$

$I_{OL} = 16 \text{ mA}$ – we need a '0' at the output of the TTL to have enough current for the LED



- Design a circuit based on a NOR TTL standard gate, able to drive a LED. For the LED, following values are considered:
 $V_{LED}=0,65V$ and $I_{LED}=20mA$.
- if $I='0'$ -> Q_1 opened -> LED on
- if $I='1'$ -> Q_1 blocked -> LED off

$$I_B = \frac{I_C}{\beta} = \frac{I_{LED}}{\beta} = 0.2mA$$

$$R_B = \frac{V_{OH\min} - V_{BE}}{I_B} = 8.25K\Omega$$

$$R_C = \frac{V_{CC} - V_{LED} - V_{CE}}{I_{LED}} = 160\Omega$$

